Team Name : MeghaMukt

Team Leader Name : B.N.V CHAITANYA YADAV

Problem Statement : PS-2 — Generative AI-Based Cloud Removal and Reconstruction for LISS-IV Satellite Imagery

Team Members

Team Leader

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Team Member 2

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College: Pragati Engineering College , Kakinada

01 HOW DIFFERENT IS IT FROM ANY OTHER IDEAS?

SAR–Guided Reconstruction
Most existing methods use only optical images. MeghaMukt fuses Sentinel-1 SAR with LISS-IV for better recovery of hidden information.

Latent Diffusion Model
Unlike traditional CNN/GAN models, we use a Latent Diffusion Model that produces more realistic, high-fidelity results with better generalization.

Spectral Fidelity
Preserves spectral signatures for analysis-ready outputs, not just visually pleasing results.

End-to-End Automated Pipeline
From data ingestion to cloud-free GeoTIFF export with quality assessment — minimal manual intervention.

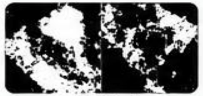
Scalable & Deployable
Built for real-world use with GPU acceleration, batch processing and API-based deployment.

02 HOW WILL IT BE ABLE TO SOLVE THE PROBLEM?

1. Data Acquisition
LISS-IV (cloudy), Sentinel-1 SAR, DEM and ancillary data

Cloudy LISS-IV

2. Preprocessing
Cloud masking (Fmask + U-Net), co-registration, radiometric correction

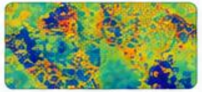


Cloud Mask

3. Data Fusion
Fuse SAR backscatter with LISS-IV to extract hidden surface information

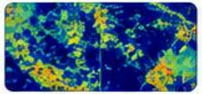
SAR Backscatter

4. AI Reconstruction
Latent Diffusion Model reconstructs cloud-covered regions using SAR guidance



Reconstruction Map

5. Quality Assessment
Evaluate using PSNR, SSIM, SAM and confidence maps



Confidence Map

6. Output Generation
Generate cloud-free, analysis-ready GeoTIFF for applications

Cloud-Free Output

03 USP OF THE PROPOSED SOLUTION

First-of-its-kind SAR-guided Latent Diffusion approach for LISS-IV cloud removal.

High-quality, spectrally consistent outputs suitable for real-world analysis.

Fully automated pipeline with integrated quality assessment.

High accuracy with better detail recovery even in thick cloud regions.

GIS-ready GeoTIFF outputs compatible with existing remote sensing workflows.

Scalable, fast and deployable for operational use by agencies.

Empowers decision-making in agriculture, disaster management, environment monitoring and infrastructure planning.

List of features offered by the solution

01

SAR-GUIDED RECONSTRUCTION

Fuses Sentinel-1 SAR with LISS-IV to accurately recover cloud-obscured regions and preserve fine details.

02

GENERATIVE AI ENGINE

Powered by Latent Diffusion Models to generate photorealistic, high-fidelity, and spatially consistent cloud-free imagery.

03

SMART CLOUD DETECTION

Advanced deep learning models (UNet + FMask) detect and precisely segment clouds before reconstruction.

04

SPECTRAL FIDELITY PRESERVATION

Maintains spectral signatures and radiometric consistency for reliable analysis and downstream applications.

05

END-TO-END AUTOMATED PIPELINE

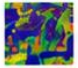
Fully automated workflow from data ingestion, prepocessing, AI reconstruction to GeoTIFF export with minimal intervention.

Cloudy Input | Ground Truth | AI Output

06

QUALITY & CONFIDENCE ANALYSIS

Evaluates output quality using PSNR, SSIM, SAM and generates confidence maps for uncertainty estimation.



- ✓ PSNR
- ✓ SSIM
- ✓ SAM
- ✓ ERGAS

07

GIS-READY GEOREFERENCED OUTPUT

Exports georeferenced GeoTIFF images compatible with all major GIS and remote sensing platforms.

GeoTIFF

08

SCALABLE & HIGH PERFORMANCE

Built for scale with GPU acceleration, batch processing and optimized inference for large-area coverage.

GPU Accelerated | Batch Processing | High Throughput | Ancillary Data Data

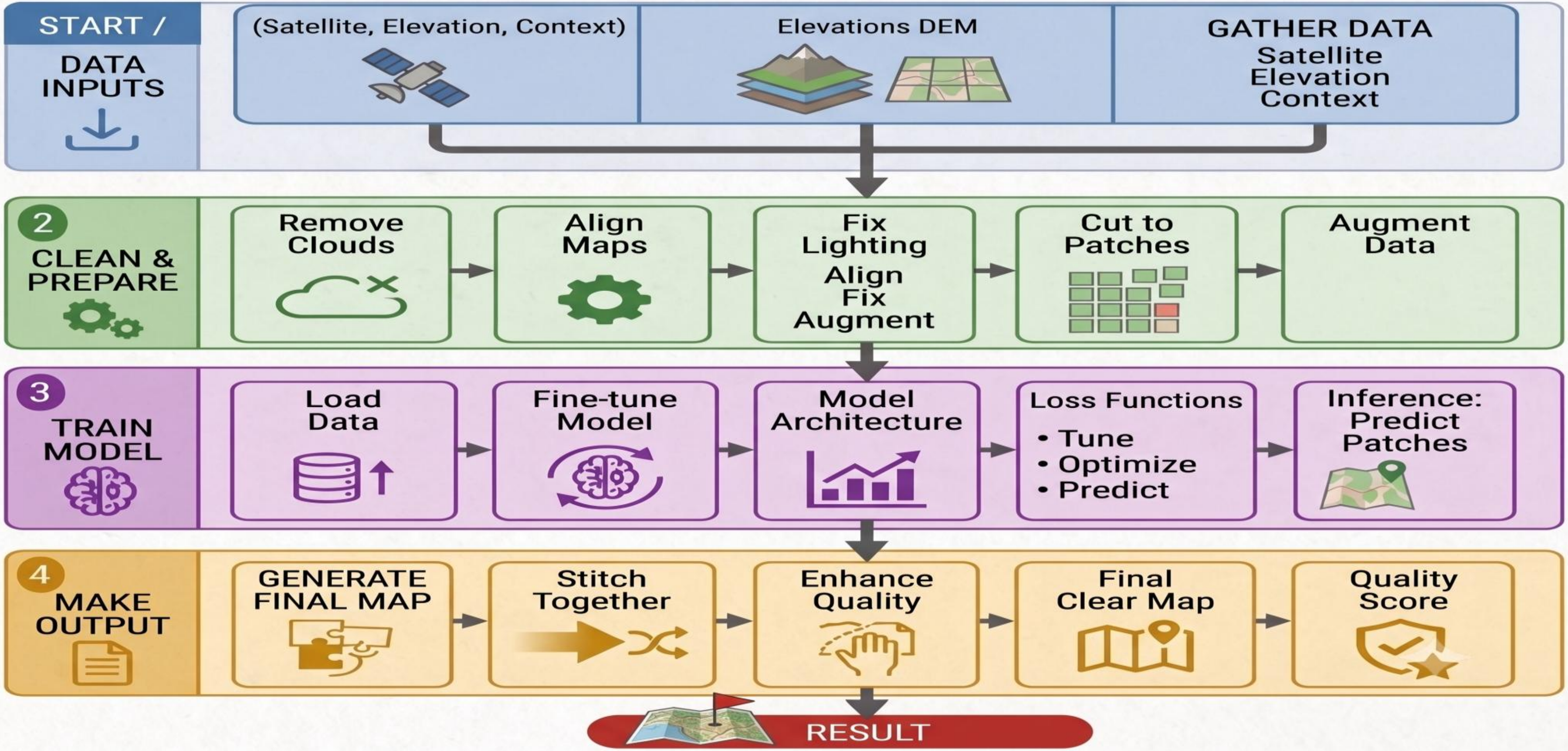
10

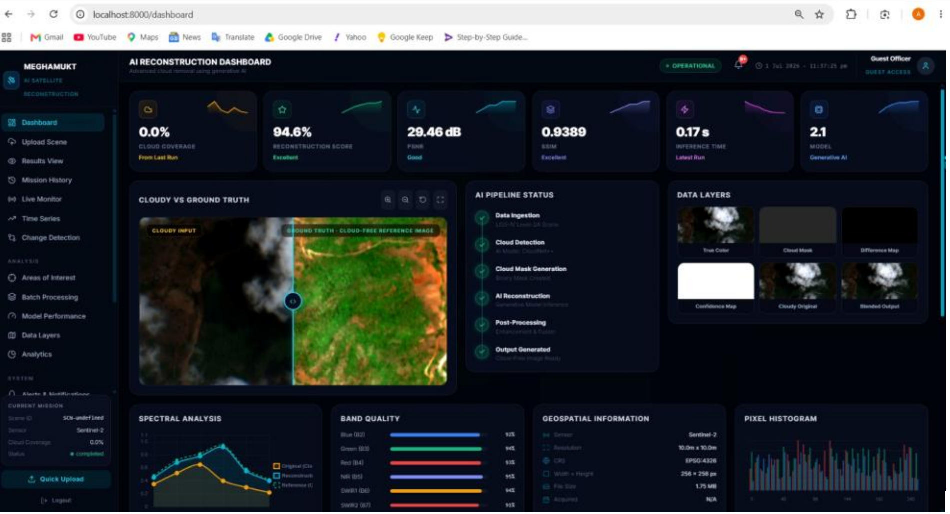
APPLICATION-DRIVEN IMPACT

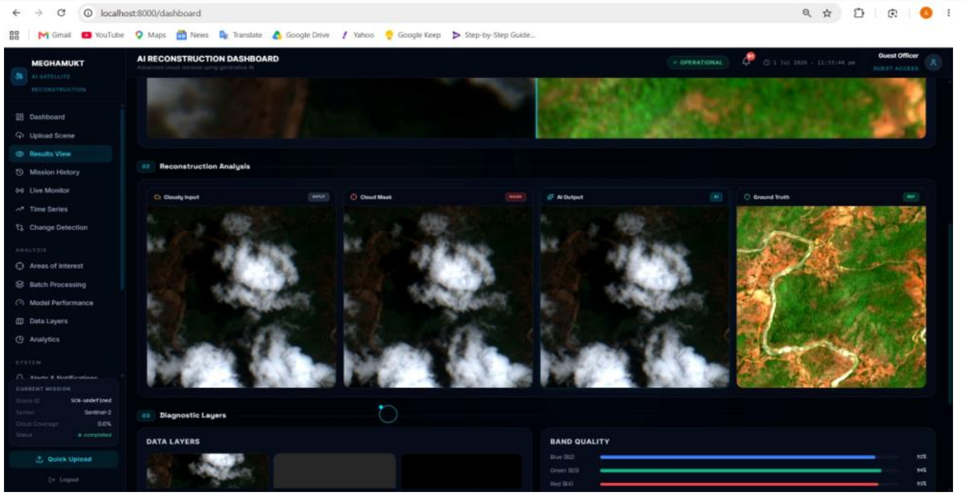
Delivers analysis-ready, cloud-free imagery that drives real-world decision making across sectors.

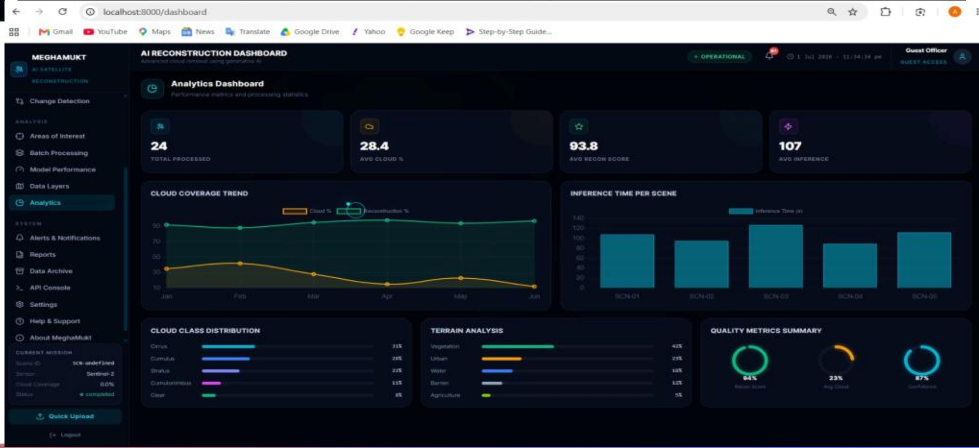
Agriculture | Disaster Mgmt. | Urban Planning | Env. Monitoring

MeghaMukt: End-to-End Workflow Flowchart





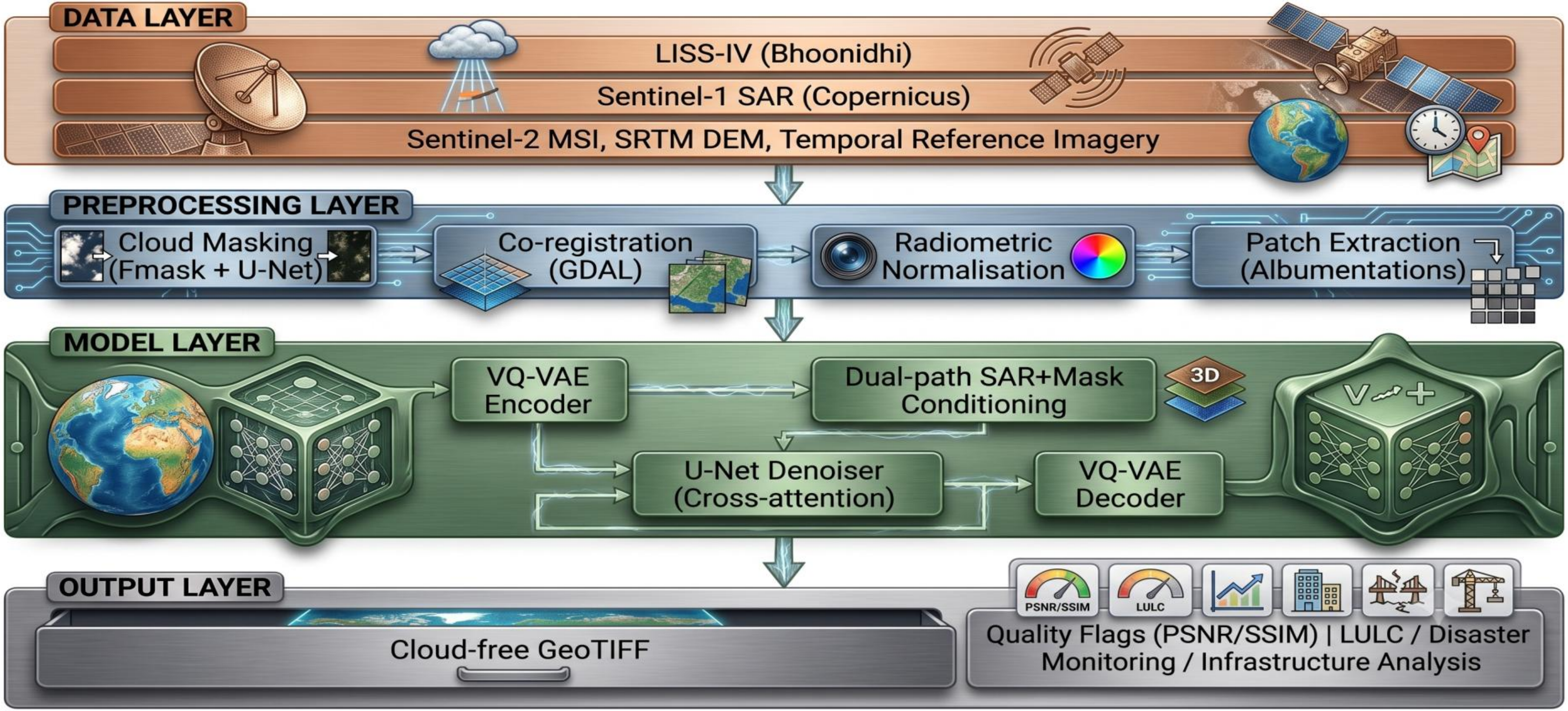




Github repository link:
https://github.com/24A31A4660/MeghaMukt

Demo Link:
https://meghamukt.vercel.app/

System Architecture



Technologies Used:

Technologies Used

01 | DEEP LEARNING & DEPLOYMENT

- Python 3.10+
- PyTorch 2.x
- NumPy (array manipulation)
- FastAPI (high-performance API)

04 | DEPLOYMENT

- Docker (containerised pipeline)
- GPU: NVIDIA A100 / V100
- GPU: NVIDIA CUDA

02 | GEOSPATIAL TOOLS & DATA SOURCES

- GDAL 3.x
- Rasterio
- QGIS (visual QA)
- Google Earth Engine
- Folium (interactive maps)
- Three.js (interactive 3D maps)

05 | DATA SOURCES

- Bhoonidhi portal (LISS-IV)
- Copernicus Open Access Hub
- SRTM / CARTOSAT DEM

03 | COMPUTER VISION, MODELS & EVALUATION

- OpenCV
- Scikit-image
- Albumentations

06 | MODELS / EVALUATION

- SatMAE (satellite imagery foundation)
- SEN12MS-CR dataset x2
- SEN12MS-CR dataset x2 (cloud removal)
- VGG-19 VGG-19 (perceptual loss)

Implementation Timeline & Cost Estimate:

Timeline (8 Weeks)

- Weeks 1–2: Data acquisition, preprocessing pipeline, cloud mask generation
- Weeks 3–4: Architecture implementation, SEN12MS-CR pretraining
- Weeks 5–6: LISS-IV fine-tuning, hyperparameter tuning, ablation studies
- Week 7: Evaluation suite, comparative benchmarking
- Week 8: Deployment API, documentation, final report

Cost Estimate (Compute)

- GPU compute (A100, 40h pretraining + 20h fine-tuning): ~₹12,000
- Data storage (250 GB for scenes + checkpoints): ~₹2,000
- Cloud platform (AWS/GCP inference deployment): ~₹4,000

Total Estimated Cost: ₹18,000 – ₹25,000

Note: Open-source tools only (PyTorch, GDAL, Rasterio). All pretrained models publicly available.

BHARATIYA ANTARIKSH HACKATHON 2026

THANK YOU